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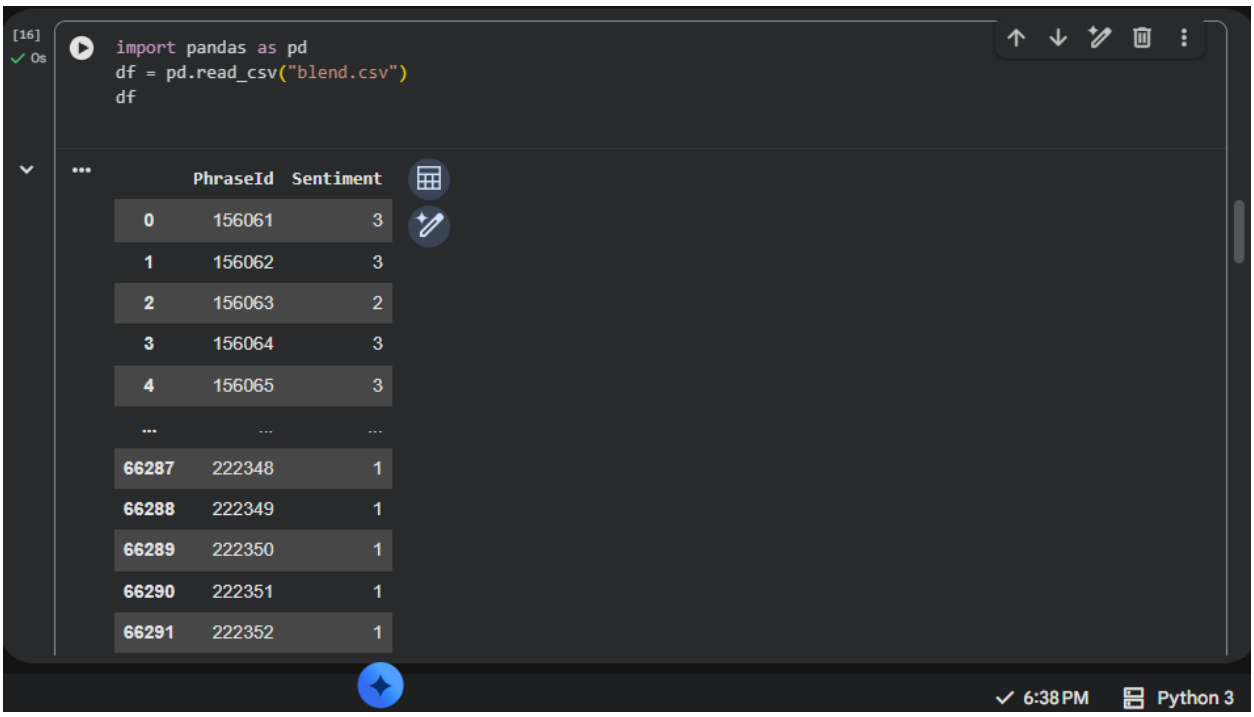
Prn = 202501080014

Dataset = Movie review

20 Problem Statement

1. Load dataset and show data



```
import pandas as pd
df = pd.read_csv("blend.csv")
df
```

	PhraseId	Sentiment
0	156061	3
1	156062	3
2	156063	2
3	156064	3
4	156065	3
...	...	...
66287	222348	1
66288	222349	1
66289	222350	1
66290	222351	1
66291	222352	1

2. First 10 rows

```
[17]
0s
print(df.head(10))

   PhraseId  Sentiment
0    156061          3
1    156062          3
2    156063          2
3    156064          3
4    156065          3
5    156066          3
6    156067          3
7    156068          2
8    156069          3
9    156070          2
```

3. Total number of reviews

```
[18]
0s
print("Total reviews:", len(df))

Total reviews: 66292
```

4. Column names

```
[20]
0s
print(df.columns)

Index(['PhraseId', 'Sentiment'], dtype='object')
```

5. Data types

```
[21]
0s
print(df.dtypes)

PhraseId     int64
Sentiment    int64
dtype: object
```

6. Missing values

```
[22]
0s
print(df.isnull().sum())

PhraseId     0
Sentiment    0
dtype: int64
```

7.Count sentiment

```
print(df['Sentiment'].value_counts())

Sentiment
2    38237
3    13550
1    11703
4     1915
0      887
Name: count, dtype: int64
```

8.Average rating

```
df.columns = df.columns.str.strip().str.lower()

print(df.columns)

Index(['phraseid', 'sentiment'], dtype='object')
```

9. Unique sentiments

```
print(df['sentiment'].unique())

[3 2 1 4 0]
```

10. Sentiment count

```
print(df['sentiment'].value_counts())

sentiment
2    38237
3    13550
1    11703
4     1915
0      887
Name: count, dtype: int64
```

s

11. Least frequent sentiment

```
print(df['sentiment'].value_counts().idxmin())
```

```
0
```

12. Total sentiment classes

```
print(df['sentiment'].nunique())
```

```
5
```

13. Percentage of each sentiment

```
print(df['sentiment'].value_counts(normalize=True) * 100)
```

```
sentiment
2    57.679660
3    20.439872
1    17.653714
4     2.888735
0     1.338020
Name: proportion, dtype: float64
```

14. Dominant sentiment

```
print(df['sentiment'].mode()[0])
```

```
2
```

15. Top 3 sentiments

```
print(df['sentiment'].value_counts().head(3))
```

```
sentiment
2    38237
3    13550
1    11703
Name: count, dtype: int64
```

16. Bottom 3 sentiments

```
print(df['sentiment'].value_counts().tail(3))
```

```
sentiment
1    11703
4     1915
0      887
Name: count, dtype: int64
```

17. Check duplicates

```
print(df.duplicated().sum())
```

```
0
```

18. Columns already cleaned (lowercase)

```
print(df.columns)
```

```
Index(['phraseid', 'sentiment', 'word_count'], dtype='object')
```

19. Random 5 rows

```
print(df.sample(5))
```

```
        phraseid  sentiment word_count
20720    176781          2   Ellipsis
44488    200549          4   Ellipsis
22524    178585          2   Ellipsis
28632    184693          2   Ellipsis
11938    167999          1   Ellipsis
```

20. Shape

```
print(df.columns[0])
```

```
phraseid
```